

Áp dụng (Apply It 2.2) - Phân tích Bán hàng (Super Scooters)



	A	B	C	D	E	F	G	H	I	J
1	OrderNumber	Model	SoldDate	SalesVolume	Color	Location	State	DaysInInventory	UnitSalePrice	GrossSales
2	13684	Celeritas	12/31/2025	20	Yellow	Dallas	TX	83	\$ 342.00	\$ 6,840
3	13685	Lazer	12/31/2025	20	Blue	Seattle	WA	52	\$ 414.00	\$ 8,280
4	13682	Captain	12/30/2025	20	Green	Seattle	WA	29	\$ 679.00	\$ 13,580
5	13683	Lazer	12/30/2025	38	Green	Miami	FL	22	\$ 376.00	\$ 14,288
3645	10957	Lazer	12/22/2023	18	Yellow	Phoenix	AZ	17	\$ 330	\$ 5,940
3646	10960	Lazer	12/31/2023	35	Red	Phoenix	AZ	24	\$ 357	\$ 12,495
3647										

Hình: Sử dụng VLOOKUP, SUM, SUMIF để tổng hợp dữ liệu giao dịch.

2.3 Tổ Chức Các Tập Dữ Liệu: Pivot Tables

PivotTable là một công cụ tương tác mạnh mẽ dùng để tóm tắt và phân tích lượng dữ liệu khổng lồ. 4 Vùng làm việc chính:



Rows (Hàng): Nhóm dữ liệu theo danh mục theo chiều dọc.



Columns (Cột): Nhóm dữ liệu theo danh mục theo chiều ngang.

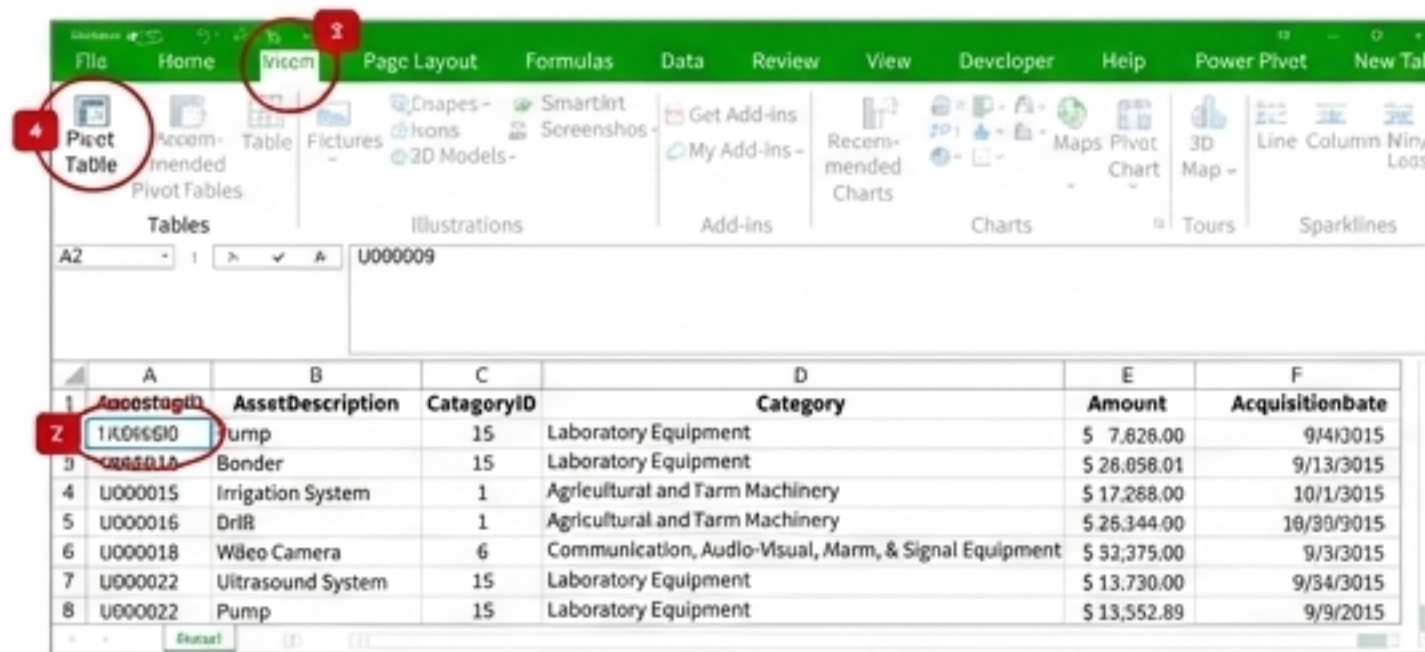


Values (Giá trị): Thực hiện tính toán (SUM, COUNT, AVERAGE,...) trên các trường số liệu.

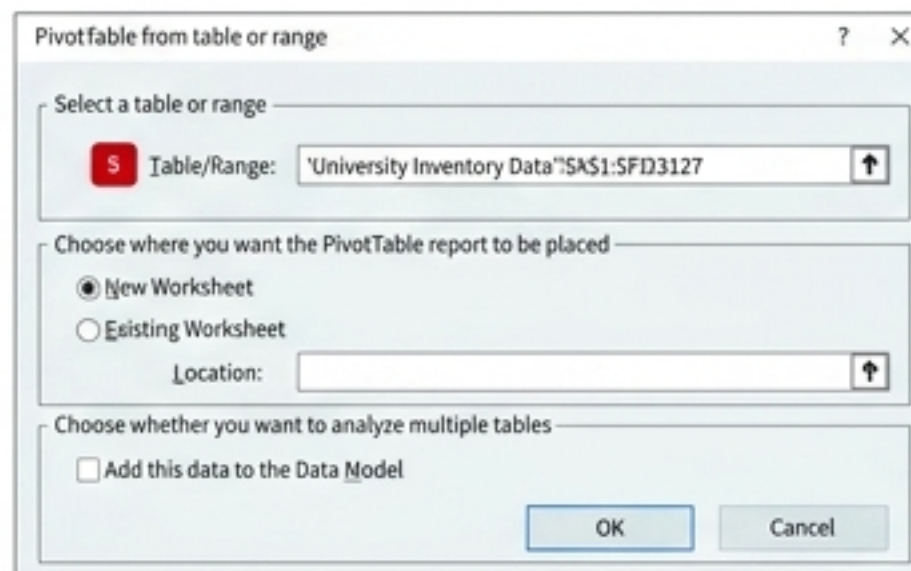


Filters (Bộ lọc): Lọc toàn bộ PivotTable theo các điều kiện.

Tạo một Microsoft Excel PivotTable

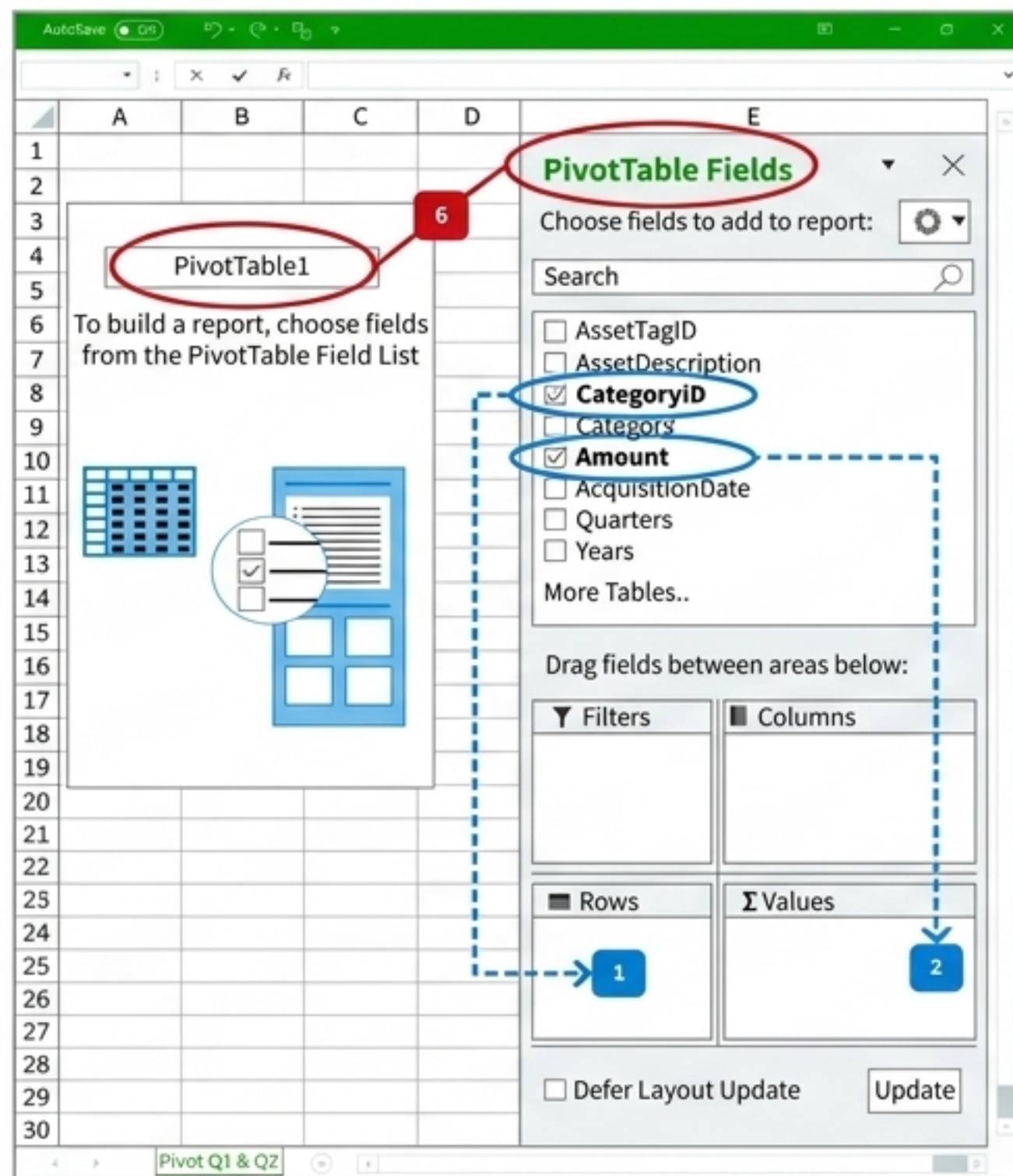


Hình: Chọn Insert > PivotTable.



Hình: Chọn vùng dữ liệu nguồn.

Vùng làm việc PivotTable (PivotTable Fields)



Hình: Danh sách các trường (Fields) và 4 vùng (Areas) kéo thả.

Trình bày Kết quả: Tổng Số dư theo Danh mục

Question 1	
Row Labels	3 Sum of Amount
Agricultural and Farm Machinery	\$ 3,838,203.58
Art Equipment	\$ 129,924.35
Athletic and Recreational Equipment	\$ 1,047,971.77
Athletic and Recreational Equipment	\$ 530,678.88
Boats, Boat Motors, and Marine Equipment	\$ 2,458,319.00
Cleaning and Maintenance Equipment	\$ 2,458,319.00
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 4,242,923.45
Computer Equipment	\$ 43,294,238.92
Drafting and Surveying Equipment	\$ 51,825.00
Electrical Appliances and Equipment	\$ 278,233.00
Electronic Equipment	\$ 1,210,703.92
Food Preparation and Serving Equipment	\$ 1,210,703.93
Furniture and Furnishings	\$ 3,060,781.37
Hand and Power Tools (Portable)	\$ 90,332.00
Industrial, Shop, and Construction Equipment	\$ 2,287,996.46
Laboratory Equipment	\$ 147,343,867.13
Material Handling Equipment	\$ 650,212.23
Medical Equipment	\$ 2,449,781.57
Miscellaneous Equipment	\$ 47,239.00
Motor Vehicles and Transportation Equipment	\$ 1,270,121.50
Musical Instruments	\$ 88,248.00
Office and Business Machines and Non-Consumable Supplies	\$ 1,587,495.45
Parking Control Equipment	\$ 230,506.72
Photographic Equipment	\$ 1,865,573.30
Plumbing, Heating, Air Conditioning, and Ventilating Equipment	\$ 146,951.15
Police, Fire Fighting, and Safety Equipment	\$ 102,458.55
Printing and Bookbinding Equipment	\$ 279,665.97
Refrigeration Equipment	\$ 6,467,000.41
Grand Total	\$ 225,069,282.74

Tìm Tổng Số dư theo Danh mục (Summarize Data)

The image shows a software interface for summarizing data. On the left, a panel titled 'Drag fields between' contains sections for 'Filters', 'Rows', and 'Columns'. The 'Rows' section has a 'Category' field with the value 'Category'. A red circle highlights the 'Value Field Settings...' button, with a red square containing the number '4' next to it. Below this button, a green box displays 'Count of Asset_T...'. At the bottom of the left panel are checkboxes for 'Defer Layout Update' and an 'Update' button.

The 'Value Field Settings' dialog box is open on the right. It has a title bar with a question mark and a close button. The 'Source Name' is 'AssetTagID'. The 'Custom Name' field contains 'Count of AssetTagID'. There are two tabs: 'Summarize Values By' and 'Show Values As'. The 'Summarize value field by' section contains a list of calculation types: Sum, Count (highlighted in blue), Average, Max, Min, and Product. At the bottom of the dialog are buttons for 'Number Format', 'OK', and 'Cancel'.

Value Field Settings

Source Name: AssetTagID

Custom Name: Count of AssetTagID

Summarize Values By Show Values As

Summarize value field by

Choose the type of calculation that you want to use to summarize data from the selected field

Sum
Count
Average
Max
Min
Product

Number Format OK Cancel

Thay đổi Phép tính (Value Field Settings)

Row Labels	Question 1	Question 2
	Sum of Amount	Count of AssetTagID
Agricultural and Farm Machinery	\$ 3,833,205.58	171
Art Equipment	\$ 129,924.35	7
Athletic and Recreational Equipment	\$ 1,047,971.77	80
Boats, Boat Motors, and Marine Equipment	\$ 530,678.88	37
Cleaning and Maintenance Equipment	\$ 2,458,319.00	182
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 4,242,923.45	271
Computer Equipment	\$ 43,294,238.92	2543
Drafting and Surveying Equipment	\$ 18,033.00	2
Electrical Appliances and Equipment	\$ 51,825.00	5
Electronic Equipment	\$ 278,233.02	29
Food Preparation and Serving Equipment	\$ 1,210,703.93	112
Furniture and Furnishings	\$ 3,060,781.37	185
Hand and Power Tools (Portable)	\$ 90,332.00	5
Industrial, Shop, and Construction Equipment	\$ 2,287,996.46	154
Laboratory Equipment	\$ 147,343,867.13	8057
Material Handling Equipment	\$ 650,212.23	47
Medical Equipment	\$ 2,449,781.57	113
Miscellaneous Equipment	\$ 47,239.00	4
Motor Vehicles and Transportation Equipment	\$ 1,270,121.50	75
Musical Instruments	\$ 88,248.00	7
Office and Business Machines and Non-Consumable Supplies	\$ 1,587,495.45	83
Parking Control Equipment	\$ 230,506.72	12
Photographic Equipment	\$ 1,865,573.30	143
Plumbing, Heating, Air Conditioning, and Ventilating Equipment	\$ 146,953.15	17
Police, Fire Fighting, and Safety Equipment	\$ 102,458.55	12
Printing and Bookbinding Equipment	\$ 279,665.57	19
Refrigeration Equipment	\$ 6,467,000.41	754
Grand Total	\$ 225,009,282.74	13126

PivotTable Fields

Choose fields to add to report

Search

☒ AssetTagID

☐ AssetDescription

☐ CategoryID

☒ Category

☒ Amount

☐ AcquisitionDate

☐ Quarters

☐ Years

More Tables..

Drag fields between areas below:

Filters

Columns

Rows

Values

Sum of Amount

Count of AssetTagID

☐ Defer Layout Update

Update

Hình: Thay đổi hàm từ Sum thành Count để đếm số lượng tài sản.

Lọc Pivot Tables (Hộp Filter)

The image shows an Excel PivotTable and its corresponding PivotTable Task Pane. The PivotTable is filtered by 'Years' (2016-2024) and sorted by 'Category'. The PivotTable Task Pane shows the 'Filters' area with 'Years' selected, the 'Rows' area with 'Category' selected, and the 'Σ Values' area with 'Sum of Amount' selected. Red annotations highlight the 'Years' filter (2), the 'Category' filter (1), and the 'Sum of Amount' filter (1).

PivotTable Data:

Category	Sum of Amount
Agricultural and Farm Machinery	\$ 258,409.17
Athletic and Recreational Equipment	\$ 8,045.00
Boats, Boat Motors, and Marine Equipment	\$ 188,366.29
Cleaning and Maintenance Equipment	\$ 103,113.50
Communication, Audio-Visual, Alarm, and Signal Equipment	\$ 23,274.20
Computer Equipment	\$ 89,889.00
Food Preparation and Serving Equipment	\$ 389,043.31
Furniture and Furnishings	
Hand and Power Tools (Portable)	
Industrial, Shop, and Construction Equipment	
Laboratory Equipment	
Material Handling Equipment	
Medical Equipment	
Miscellaneous Equipment	
Motor Vehicles and Transportation Equipment	
Office and Business Machines and Non-Consumable Supplies	
Parking Control Equipment	
Printing and Bookbinding Equipment	
Refrigeration Equipment	
Grand Total	\$ 20,869,916.66

PivotTable Task Pane:

- Filters:** Years
- Rows:** Category
- Σ Values:** Sum of Amount

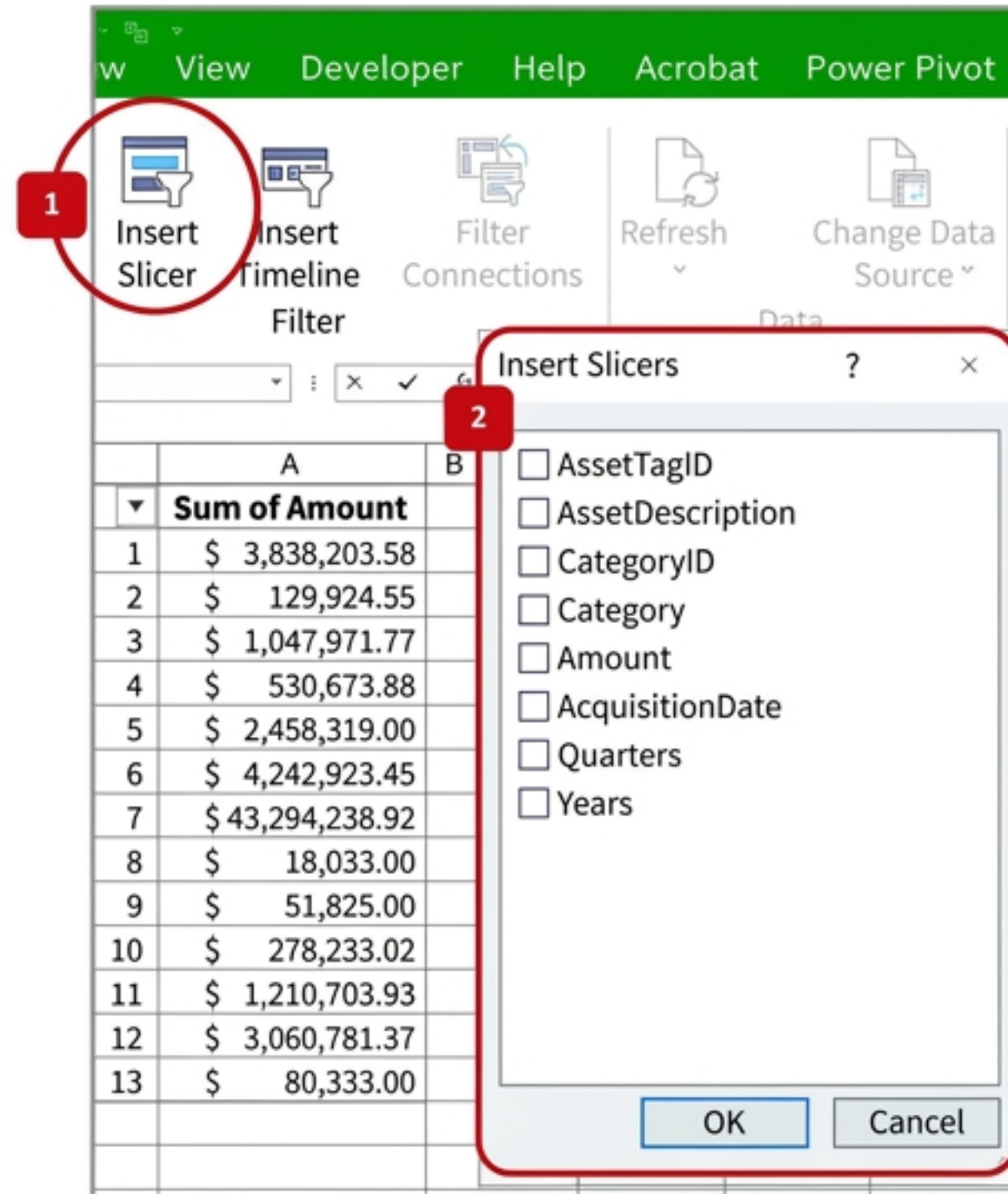
☐ Select Multiple Items

☐ Defer Layout update

Update

Hình: Kéo trường Asset Sub-Category vào vùng Filters để tạo bộ lọc tổng.

Adding Slicers to PivotTables



Sử dụng Slicers để Lọc Trực quan

The image shows the Power BI ribbon interface with three main panes: Filters, Visualizations, and Fields.

- Filters Pane:** Contains a search bar and three filter sections: "Filters on this visual", "Filters on this page", and "Filters on this page". Each section has a search bar and a button labeled "Add data fields here".
- Visualizations Pane:** Contains a grid of visualization icons. The Slicer icon (a vertical bar chart) is circled in red. Below the grid is a "Values" section with a list of fields: "Category" and "Amount". Below the "Values" section is a "Drill through" section with a toggle switch set to "Off" and a button labeled "Keep all filters".
- Fields Pane:** Contains a search bar and a list of fields under the "Sheet1" tab. The fields are: "AcquisitionDate", "Amount" (checked), "AssetDescription", "AssetTagID", "Category" (checked), "CategoryID" (checked), and "Year".

Áp dụng (Apply It 2.3) - Phân tích Bán hàng bằng PivotTables

Year	2025
Row Labels	Sum of Gross Sales
Boston	\$ 1,433,578.00
Charlotte	\$ 1,653,506.00
Chicago	\$ 1,624,719.00
Dallas	\$ 1,134,785.00
Miami	\$ 1,606,576.00
Phoenix	\$ 1,376,503.00
Salt Lake City	\$ 1,593,184.00
Seattle	\$ 1,742,311.00
Grand Total	\$ 12,165,162.00

Hình: Tạo báo cáo doanh số theo tháng và sản phẩm.

2.4 Thước Đo Mô Tả Giúp Hiểu Dữ Liệu

Thống kê mô tả được sử dụng trong giai đoạn Khám phá Dữ liệu (Giai đoạn 2). 3 nhóm thước đo chính:



**Vị trí trung tâm
(Location/Central
Tendency):** Mean
(Trung bình), Median
(Trung vị), Mode (Yếu vị).

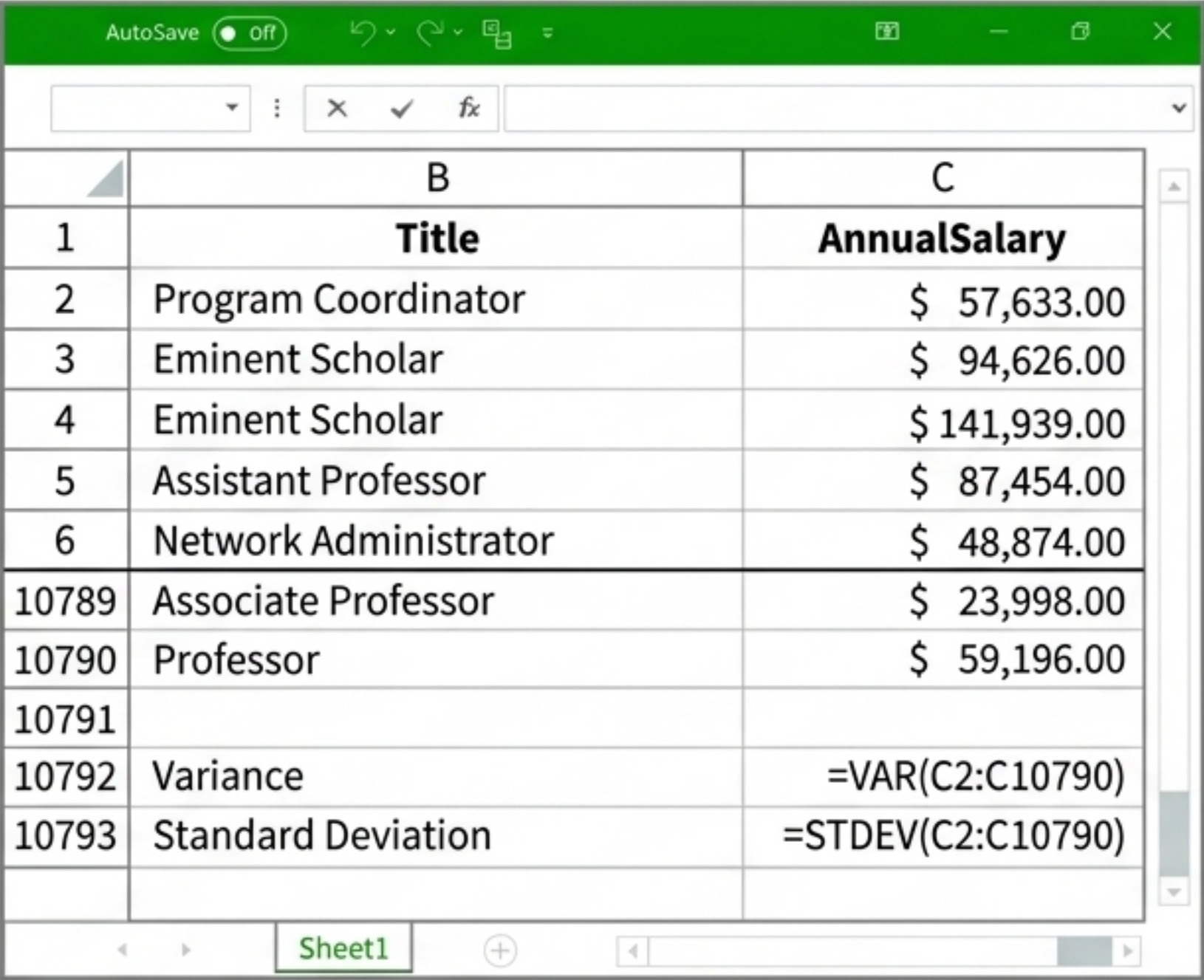


**Độ phân tán
(Dispersion):** Range
(Khoảng), Variance
(Phương sai), Standard
Deviation (Độ lệch
chuẩn).



Hình dạng (Shape):
Skewness (Độ lệch),
Kurtosis (Độ nhọn).

Tính toán Các Thước đo Vị trí (Mean, Median, Mode)

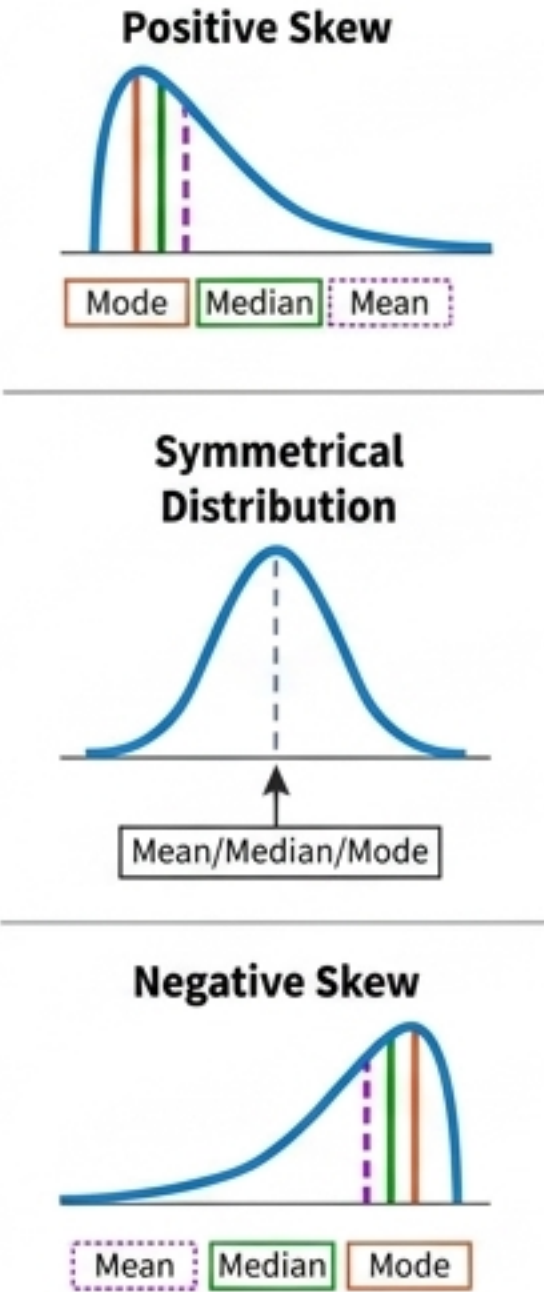


The screenshot shows an Excel spreadsheet with the following data:

	B	C
1	Title	AnnualSalary
2	Program Coordinator	\$ 57,633.00
3	Eminent Scholar	\$ 94,626.00
4	Eminent Scholar	\$ 141,939.00
5	Assistant Professor	\$ 87,454.00
6	Network Administrator	\$ 48,874.00
10789	Associate Professor	\$ 23,998.00
10790	Professor	\$ 59,196.00
10791		
10792	Variance	=VAR(C2:C10790)
10793	Standard Deviation	=STDEV(C2:C10790)

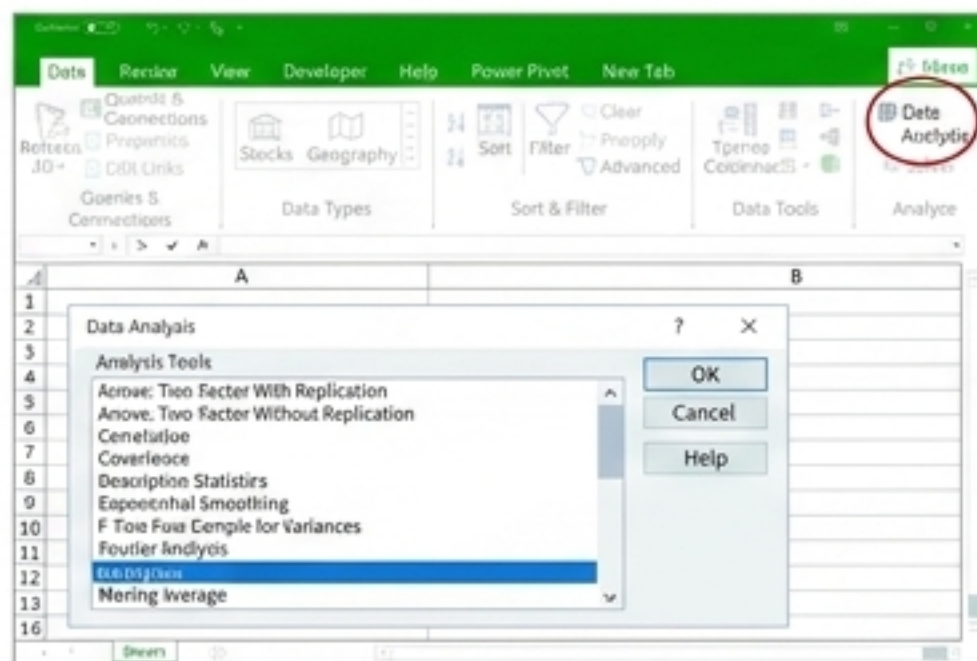
Hình: Sử dụng hàm AVERAGE, MEDIAN, MODE.SNGL trong Excel.

Các thước đo Hình dạng (Skewness)

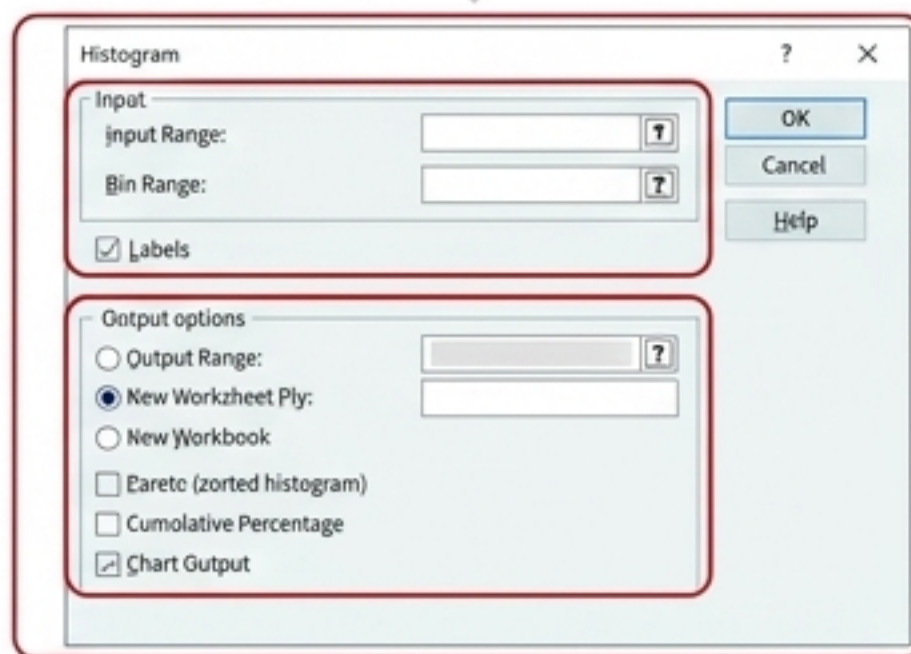


Hình: Phân phối lệch trái (Left-Skewed) và lệch phải (Right-Skewed).

Thiết lập Công cụ Phân tích Dữ liệu (Data Analysis Tool)



Các Công cụ Thống kê Mô tả Excel (Data Analysis Toolpak) - Hình: Tự động xuất toàn bộ các chỉ số thống kê cơ bản trong 1 cú nhấp chuột.



Biểu đồ Tần suất (Histograms) - Hình: Xây dựng bảng phân phối tần số (Bins).

Biểu đồ Tần suất & Phân tích Tương quan

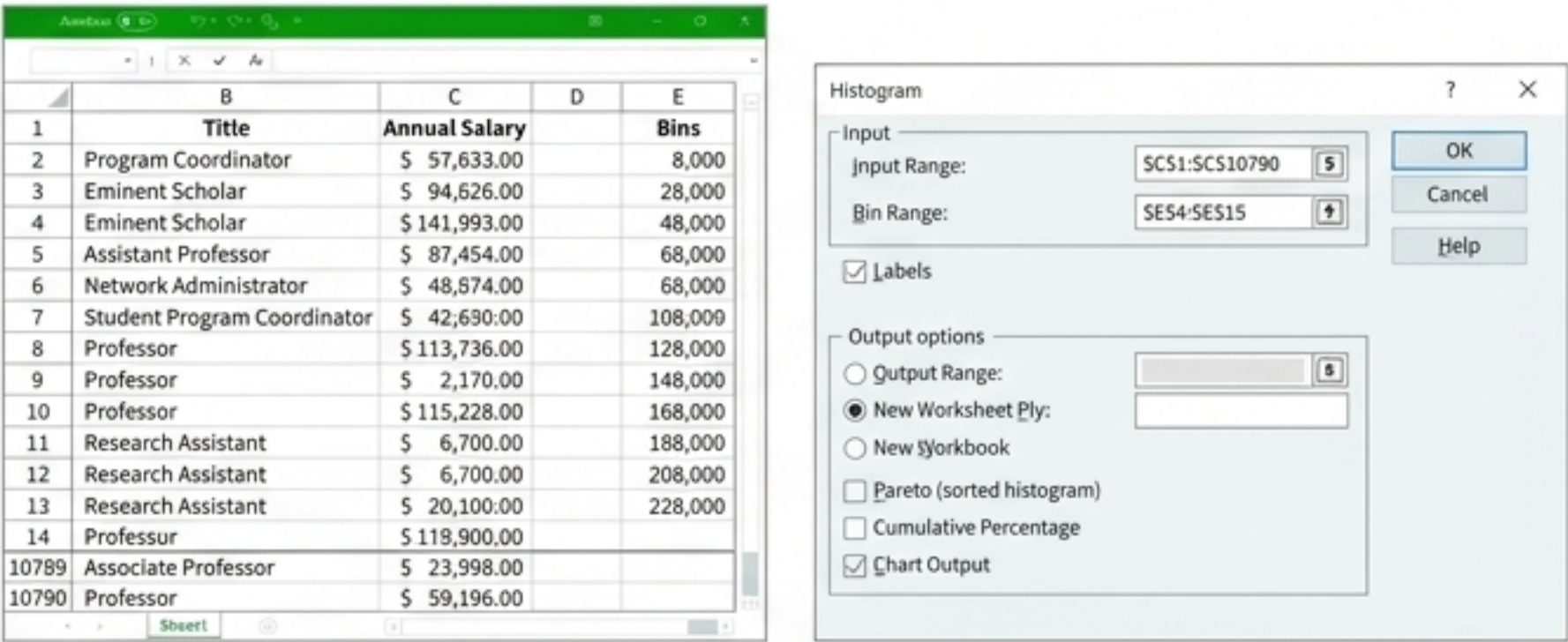
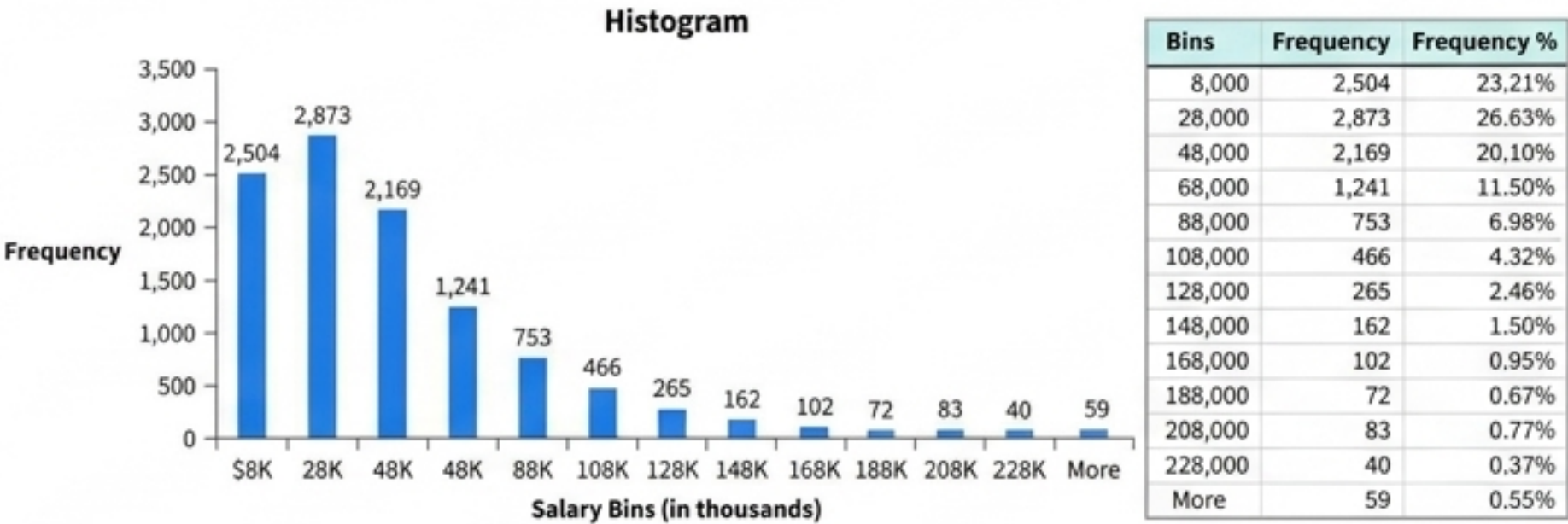


ILLUSTRATION 2.46 University Payroll Data Histogram and Frequency Distribution



Hình: Trực quan hóa bằng Histogram.